Raid Forge — Technical Sheet

Tech used

Single self-contained HTML file (about 27 KB), no libraries or external assets. HTML5 Canvas 2D with devicePixelRatio scaling (capped at 2x): extruded neon buildings with glow, rotating turrets, a pulsing Core crystal, flickering hut windows, cannon tracers, chained lightning, mortar arcs, particle bursts, rubble, floating gold numbers, screenshake and a twinkling starfield.

Web Audio API: every sound is synthesized (per-defense shots, explosions, sapper blasts, deploys, star fanfare) plus a procedural battle track with kick, snare, bass and lead that speeds up and opens its filter as the timer runs down or the Core falls. Audio starts on first tap (iOS-safe), can be muted, and suspends when the tab is hidden.

Pointer Events with tap-and-hold deploying, 70px troop buttons, safe-area insets, a board that scales to any portrait screen. Saves in localStorage.

Game systems

12x16 grid. Bases are generated from a seed per level: a 2x2 Core, a walled ring with one gap from level 2, defenses unlocked by level, loot buildings and huts, extra wall pockets from level 9. Building health grows 7% per level.

4 troop types with targeting rules (nearest, defenses only, walls only), wall blocking, ranged and suicide splash attacks. 4 defense types: single-target, fast, splash with a dead zone, and chain lightning.

Stars, 90-second timer, army capacity, gold loot, 5 upgrade tracks with 10 levels each, 20-base campaign map with best stars.

Audit 1 — Balance (simulated play)

A headless simulator runs the same game code as the HTML file. Three bot styles each played 6 full campaigns (up to 80 raids, buying the cheapest upgrade whenever they can afford it): a weak bot that drops only Brutes in one spot, a mixed bot that spreads Brutes and Archers, and a "smart" bot that sends Sappers and Rams first in a tight group.

First pass: building health grew 9% per level and the smart bot stalled at base 6 of 20. Tuned to 7% per level and raised Ram health from 340 to 420.

Final results: weak: cleared 17.8/20 bases in 79 raids avg; mixed: cleared 20.0/20 bases in 29 raids avg; smart: cleared 19.0/20 bases in 80 raids avg; smart, no upgrades, 1st-try wins L1:10/10 L5:0/10 L9:0/10 L13:0/10 L17:0/10. Spreading out beats clumping because of splash defenses, which is the lesson the game teaches. Later bases need upgrades, so replaying for gold is part of the loop.

Audit 2 — Bugs

Ran a headless smoke test of the real file with a stubbed browser: menu, level select, taps outside the drop zone, deploying, a full raid to 3 stars, gold and unlocks, the Armory, retry, map, sound toggle, a forced defeat on base 20 and saving. No errors.

Fixed: a troop already killed by a splash or chain hit in the same shot could trigger a second death effect and sound.

Checked: the raid ends as soon as the army is spent and no troops remain, the 90-second timer always ends a stuck raid, and the end screen can only appear once.

Audit 3 — Player experience

First raid shows "Tap the green zone to drop troops" on the map and a hint toast. Each raid opens with a toast listing the defenses you face.

Tapping outside the zone, running out of army or not having enough for a Ram explains why nothing happened. Tapping a defense shows its range circle.

The timer turns red in the last 10 seconds. A failed raid tells you to visit the Armory or spread your troops.